XRPL Validator — Daily Engineering Report · 2026-08-01

Period: 2026-08-01 00:00 to 24:00 US Eastern (2026-08-01 04:00 to 2026-08-02 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-08-01): 0.99758 (36 missed of 14,861). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): consolidate NFToken pages when a removal shrinks one** (`761df02`, xrpl-validator) `page_remove` handled an EMPTIED page (delete, relink neighbours) but did nothing when a page merely shrank. rippled consolidates on every removal (NFTokenHelpers.cpp removeToken): "The current page isn't empty. Update it and then try to consolidate pages. Note that this consolidation attempt may actually merge three pages into one!"
- **fix(ledger): split a full NFToken page before inserting** (`9a82325`, xrpl-validator) `page_insert` had NO capacity check: it located the page and inserted, so a full page silently grew to 33 tokens — past kDirMaxTokensPerPage. rippled splits first (NFTokenHelpers.cpp getPageForToken): the lower part moves to a NEW page keyed below, the rest stays, and the predecessor is relinked. That Created + Modified pair is exactly what mainnet emits.
- **fix(probe): hydrate trust lines for the currencies named in Paths** (`549abb1`, xrpl-validator) We hydrated only the Amount and SendMax currencies, never the intermediate hops. A payment that ripples through a third currency therefore saw NO trust line for that hop, ran trustCreate for a line that already exists, and invented a directory page to hold it.
- **fix(ledger): align IOU addition the way STAmount does** (`d5a08c8`, xrpl-validator) `STAmount operator+` brings both operands to the LARGER exponent by integer division: while (ov1 < ov2) { vv1 /= 10; ++ov1; } while (ov2 < ov1) { vv2 /= 10; ++ov2; }
- **fix(ledger): apply tfSetfAuth to the trust line** (`4bac28e`, xrpl-validator) We honoured tfSetfAuth in preclaim's redundancy test but never APPLIED it. `do_apply` handled tfSetNoRipple / tfClearNoRipple / tfSetFreeze /

tfClearFreeze and had no branch for the auth bit, so a TrustSet whose only effect is authorisation left the line untouched entirely.

- **fix(ledger): CredentialCreate joins both owner directories** (`be244f0`, xrpl-validator) A credential is inserted into the issuer's AND the subject's owner directory, but is owned only by the issuer — CredentialCreate.cpp:147-174: "Added to both dirs, owned only by issuer. CredentialAccept will transfer ownership to subject." So the issuer's dirInsert carries the single adjustOwnerCount(+1) and the subject's adds no reserve at all. A self-issued credential (subject == account) skips the subject insert …

Independence program

No independence record for UTC day 2026-08-01; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Credential-Gated Auto-Preauth for Tokens** (compliance) Add an optional sfAllowedCredentials SField (STArray, max 8 entries of {Issuer, CredentialType}) to MPTokenIssuance, usable only when lsfMPTRequireAut
- **Native NFT Loan Delegation Without Remint** (nft) Add two SFields to NFTokenPage entries, sfBorrower (AccountID) and sfLoanExpiration (LedgerIndex), settable only via a new NFTokenLend transaction (ow
- **Deterministic Scheduled & Oracle-Triggered Execution** (hooks) Add a new owner-reserved ledger object ltSCHEDULED_TX (keylet::scheduledTx(account, sequence)) created by ScheduledTxCreate, holding a pre-signed inne

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-01 05:56	—	xrpl-validator	`761df02`	fix(ledger): consolidate NFToken pages when a removal shrinks one
08-01 18:07	—	xrpl-validator	`9a82325`	fix(ledger): split a full NFToken page before inserting
08-01 19:26	—	xrpl-validator	`549abb1`	fix(probe): hydrate trust lines for the currencies named in Paths
08-01 20:16	—	xrpl-validator	`d5a08c8`	fix(ledger): align IOU addition the way STAmount does
08-02 03:08	—	xrpl-validator	`4bac28e`	fix(ledger): apply tfSetfAuth to the trust line
08-02 03:52	—	xrpl-validator	`be244f0`	fix(ledger): CredentialCreate joins both owner directories

Times in UTC. A commit appears on the day it was committed to main or merged into it.